Instituto de Síntesis Química y Catálisis Homogénea

INTRODUCTION TO CHEMINFORMATICS

IAQCat Formation

April 28th | 2026

CHEMINFORMATICS

Combines chemistry, computer science, and information science to organize, process, and present chemical data.

Computers don't know chemistry, the translation of "chemistry" to computer language is necessary.

SMILES

It means **S**implified **M**olecular **I**nput **L**ine **E**ntry **S**ystem.

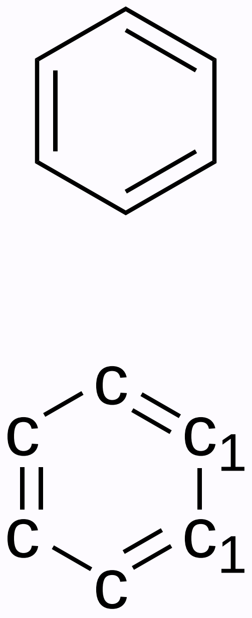
A way to write molecules in one string that is easy for humans to read.

Examples:

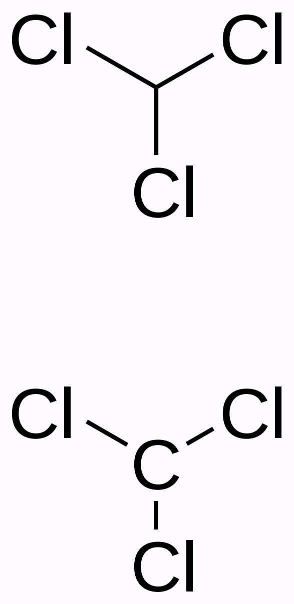
CCC

CC=C

CC#C



c1ccccc1

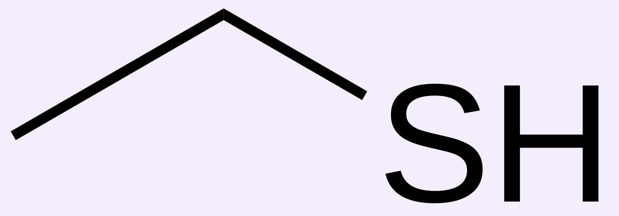


ClC(Cl)Cl

Tutorial on SMILES for more information.

CANONICAL SMILES

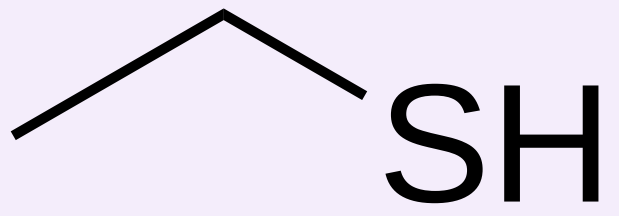
SMILES OF MOLECULE?



a) CCS
b) C(S)C
c) SCC

CANONICAL SMILES

SMILES OF MOLECULE?



a) CCS
b) C(S)C
c) SCC

ANSWER

All are correct, but in a database, will lead to **duplicates**.

The canonical SMILES must be obtained to get a unique standardized text representation:

CCS

You can check with this tool.

HANDS ON TIME

Open the Introduction to Cheminformatics jupyter notebook